

## Qualification Pack

# Data Analysis Assistant

QP Code: NIE/ITS/Q1103

Version: 1.0

NSQF Level: 3

National Institute of Electronics and Information Technology (NIELIT) || NIELIT Bhawan, Plot No. 3, PSP  
Pocket, Sector-8, Dwarka  
New Delhi-110077 || email:puneetnema@nielit.gov.in

## Qualification Pack

## Contents

|                                                                        |    |
|------------------------------------------------------------------------|----|
| NIE/ITS/Q1103: Data Analysis Assistant .....                           | 3  |
| <i>Brief Job Description</i> .....                                     | 3  |
| Applicable National Occupational Standards (NOS) .....                 | 3  |
| <i>Compulsory NOS</i> .....                                            | 3  |
| <i>Qualification Pack (QP) Parameters</i> .....                        | 3  |
| DGT/VSQ/N0101: Employability Skills (30 Hours) .....                   | 5  |
| NIE/ITS/N1116: Implementation of Data Analysis Assistant Project ..... | 11 |
| NIE/ITS/N1105: Visualize data graphically .....                        | 16 |
| NIE/ITS/N1105: Perform basic calculation using spreadsheet- .....      | 20 |
| NIE/ITS/N1109: Implementation of use cases of Data Analytics .....     | 24 |
| NIE/ITS/N1109: Manage structured data- .....                           | 28 |
| NIE/ITS/N1112: Analyze data using spreadsheet tool- .....              | 32 |
| NIE/ITS/N1113: Manage data in Open source tool- .....                  | 36 |
| NIE/ITS/N1114: Visualize Data Graphically .....                        | 40 |
| NIE/ITS/N1115: Implementation of use cases of Data Analytics- .....    | 44 |
| Assessment Guidelines and Weightage .....                              | 47 |
| <i>Assessment Guidelines</i> .....                                     | 47 |
| <i>Assessment Weightage</i> .....                                      | 48 |
| Acronyms .....                                                         | 50 |
| Glossary .....                                                         | 51 |

## Qualification Pack

### NIE/ITS/Q1103: Data Analysis Assistant

#### Brief Job Description

The Data Analyst Assistant Qualification is meant to inculcate the basic skills of data handling, processing and visualization. The trained learner will be able to view data in summary and aggregated form, he/she would be able to formulate equation for prediction.

#### Personal Attributes

should possess a strong analytical mindset, attention to detail, and problem-solving skills to interpret and process data effectively. Proficiency in statistical tools, data visualization software, and programming languages like Python or SQL is essential. Excellent communication skills are necessary to present findings clearly to stakeholders. Additionally, a keen eye for patterns, logical reasoning, and adaptability to new technologies are crucial for success in this role. Time management and the ability to work both independently and collaboratively enhance efficiency in handling complex datasets and delivering meaningful insights.

#### Applicable National Occupational Standards (NOS)

##### Compulsory NOS:

1. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
2. [NIE/ITS/N1116: Implementation of Data Analysis Assistant Project](#)
3. [NIE/ITS/N1105: Perform basic calculation using spreadsheet-](#)
4. [NIE/ITS/N1109: Manage structured data-](#)
5. [NIE/ITS/N1112: Analyze data using spreadsheet tool-](#)
6. [NIE/ITS/N1113: Manage data in Open source tool-](#)
7. [NIE/ITS/N1114: Visualize Data Graphically](#)
8. [NIE/ITS/N1115: Implementation of use cases of Data Analytics-](#)

#### Qualification Pack (QP) Parameters

|                   |                               |
|-------------------|-------------------------------|
| <b>Sector</b>     | Information Technology Sector |
| <b>Sub-Sector</b> | BIOINFORMATICS                |

### Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Occupation</b>                                         | DATA ANALYTICS                                                                                                                                                                                                                                                                                      |
| <b>Country</b>                                            | India                                                                                                                                                                                                                                                                                               |
| <b>NSQF Level</b>                                         | 3                                                                                                                                                                                                                                                                                                   |
| <b>Credits</b>                                            | 10                                                                                                                                                                                                                                                                                                  |
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | 2511.0105                                                                                                                                                                                                                                                                                           |
| <b>Minimum Educational Qualification &amp; Experience</b> | 10th grade pass with NA of experience<br>OR<br>Grade 8 pass with 2 years of (NTC/ NAC) after 8th with NA of experience<br>OR<br>8th grade pass and pursuing continuous schooling in regular school with vocational subject with NA of experience<br>OR<br>8th grade pass with 2 Years of experience |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                                     |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                  |
| <b>Minimum Job Entry Age</b>                              | 16 Years                                                                                                                                                                                                                                                                                            |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                  |
| <b>Next Review Date</b>                                   | 29/03/2026                                                                                                                                                                                                                                                                                          |
| <b>NSQC Approval Date</b>                                 | 29/03/2023                                                                                                                                                                                                                                                                                          |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                 |
| <b>Reference code on NQR</b>                              | QG-03-IT-00347-2023-V1-NIELIT                                                                                                                                                                                                                                                                       |
| <b>NQR Version</b>                                        | 1                                                                                                                                                                                                                                                                                                   |

## Qualification Pack

### DGT/VSQ/N0101: Employability Skills (30 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

**PC1.** understand the significance of employability skills in meeting the job requirements

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

**PC2.** identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

**PC3.** explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

##### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

**PC4.** speak with others using some basic English phrases or sentences

##### *Communication Skills*

To be competent, the user/individual on the job must be able to:

**PC5.** follow good manners while communicating with others

**PC6.** work with others in a team

## Qualification Pack

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

**PC7.** communicate and behave appropriately with all genders and PwD

**PC8.** report any issues related to sexual harassment

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

**PC9.** use various financial products and services safely and securely

**PC10.** calculate income, expenses, savings etc.

**PC11.** approach the concerned authorities for any exploitation as per legal rights and laws

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

**PC12.** operate digital devices and use its features and applications securely and safely

**PC13.** use internet and social media platforms securely and safely

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

**PC14.** identify and assess opportunities for potential business

**PC15.** identify sources for arranging money and associated financial and legal challenges

### *Customer Service*

To be competent, the user/individual on the job must be able to:

**PC16.** identify different types of customers

**PC17.** identify customer needs and address them appropriately

**PC18.** follow appropriate hygiene and grooming standards

### *Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

**PC19.** create a basic biodata

**PC20.** search for suitable jobs and apply

**PC21.** identify and register apprenticeship opportunities as per requirement

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** need for employability skills

**KU2.** various constitutional and personal values

**KU3.** different environmentally sustainable practices and their importance

**KU4.** Twenty first (21st) century skills and their importance

**KU5.** how to use basic spoken English language

**KU6.** Do and dont of effective communication

**KU7.** inclusivity and its importance

**KU8.** different types of disabilities and appropriate communication and behaviour towards PwD

**KU9.** different types of financial products and services

## Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the job requirements                                                                                                                                                                | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                     | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC2.</b> identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                             | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC3.</b> explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc. | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                    | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC4.</b> speak with others using some basic English phrases or sentences                                                                                                                                                                                    | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                                                                                                                    | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC5.</b> follow good manners while communicating with others                                                                                                                                                                                                | -            | -               | -             | -          |
| <b>PC6.</b> work with others in a team                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                                                                                                               | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC7.</b> communicate and behave appropriately with all genders and PwD                                                                                                                                                                                      | -            | -               | -             | -          |
| <b>PC8.</b> report any issues related to sexual harassment                                                                                                                                                                                                     | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                                                                                                            | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC9.</b> use various financial products and services safely and securely                                                                                                                                                                                    | -            | -               | -             | -          |



### Qualification Pack

| Assessment Criteria for Outcomes                                                                  | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> calculate income, expenses, savings etc.                                             | -            | -               | -             | -          |
| <b>PC11.</b> approach the concerned authorities for any exploitation as per legal rights and laws | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                   | <b>4</b>     | <b>6</b>        | -             | -          |
| <b>PC12.</b> operate digital devices and use its features and applications securely and safely    | -            | -               | -             | -          |
| <b>PC13.</b> use internet and social media platforms securely and safely                          | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                           | <b>3</b>     | <b>5</b>        | -             | -          |
| <b>PC14.</b> identify and assess opportunities for potential business                             | -            | -               | -             | -          |
| <b>PC15.</b> identify sources for arranging money and associated financial and legal challenges   | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                           | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC16.</b> identify different types of customers                                                | -            | -               | -             | -          |
| <b>PC17.</b> identify customer needs and address them appropriately                               | -            | -               | -             | -          |
| <b>PC18.</b> follow appropriate hygiene and grooming standards                                    | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC19.</b> create a basic biodata                                                               | -            | -               | -             | -          |
| <b>PC20.</b> search for suitable jobs and apply                                                   | -            | -               | -             | -          |
| <b>PC21.</b> identify and register apprenticeship opportunities as per requirement                | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                  | <b>20</b>    | <b>30</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0101                   |
| <b>NOS Name</b>            | Employability Skills (30 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 2                               |
| <b>Credits</b>             | 1                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 06/02/2026                      |
| <b>Next Review Date</b>    | 06/02/2029                      |
| <b>NSQC Clearance Date</b> | 06/02/2026                      |

## Qualification Pack

### NIE/ITS/N1116: Implementation of Data Analysis Assistant Project

#### Description

The Data Analysis Assistant syllabus covers essential skills required for handling and interpreting data. It includes topics such as spreadsheet calculations, database management, data analysis with statistical tools, and data visualization using Tableau. Students will learn fundamental database concepts, SQL, and open-source tools like Linux. The course also emphasizes employability skills and hands-on project work, ensuring practical experience in real-world data analytics applications. With a total duration of 300 hours, this program prepares students for entry-level roles in data analysis by equipping them with technical and analytical skills.

#### Scope

The scope covers the following :

- The scope includes performing calculations using spreadsheets, managing structured data with databases, analyzing data using statistical techniques, and utilizing open-source tools like Linux. Additionally, it covers data visualization using Tableau and implementing real-world data analytics use cases. Students will develop practical skills in data manipulation, SQL queries, pivot tables, statistical analysis, and dashboard creation. The project will enhance analytical capabilities and job readiness through structured technical learning and hands-on applications.

#### Elements and Performance Criteria

##### *Data Processing and Calculation*

To be competent, the user/individual on the job must be able to:

- PC1.** Apply basic and advanced spreadsheet functions for data manipulation.
- PC2.** Perform sorting, filtering, and conditional formatting for data structuring.
- PC3.** Utilize pivot tables and "What-If" analysis for insights.

##### *Database Management*

To be competent, the user/individual on the job must be able to:

- PC4.** Understand database concepts and implement SQL queries for data retrieval
- PC5.** Ensure data integrity using keys and constraints.
- PC6.** Perform data migration and aggregation operations.

##### *Statistical Analysis and Data Interpretation*

To be competent, the user/individual on the job must be able to:

- PC7.** Implement statistical techniques, including probability and regression analysis.
- PC8.** Conduct descriptive analytics for trend identification.
- PC9.** Utilize statistical models for decision-making.

##### *Open-Source Data Management*

To be competent, the user/individual on the job must be able to:

- PC10.** Work with Linux-based systems for file management and command-line operations.
- PC11.** Utilize scripting for data handling and automation.

## Qualification Pack

### *Data Visualization and Reporting*

To be competent, the user/individual on the job must be able to:

- PC12.** Use Tableau to create interactive dashboards and reports.
- PC13.** Connect and analyze data from multiple sources.
- PC14.** Generate graphical representations to enhance data storytelling.

### *Implementation of Real-World Use Cases*

To be competent, the user/individual on the job must be able to:

- PC15.** Apply acquired skills to industry-relevant datasets.
- PC16.** Develop reports with actionable insights.
- PC17.** Present findings using visual and statistical data interpretations.

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Covers fundamental data analysis concepts, data structures, statistical methods, and principles of data ethics, security, and confidentiality.
- KU2.** Includes proficiency in spreadsheets, SQL for data management, Linux-based operations, Tableau for visualization, and basic Python/R for data analytics.
- KU3.** Focuses on the role of data analysis in business decision-making, industry compliance requirements, data-driven communication, teamwork, and operational efficiency improvements.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to interpret data reports, document findings clearly, and communicate insights effectively to stakeholders.
- GS2.** Applying data-driven decision-making strategies to solve business problems and optimize operations.
- GS3.** Managing datasets efficiently, structuring workflows, and ensuring data integrity.
- GS4.** Understanding user requirements and presenting data insights that drive customer-focused improvements.
- GS5.** Identifying data discrepancies, troubleshooting errors, and ensuring accurate analysis.
- GS6.** valuating data trends, identifying patterns, and drawing meaningful conclusions to support business strategies.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Data Processing and Calculation</i>                                                       | -            | -               | -             | <b>30</b>  |
| <b>PC1.</b> Apply basic and advanced spreadsheet functions for data manipulation.            | -            | -               | -             | -          |
| <b>PC2.</b> Perform sorting, filtering, and conditional formatting for data structuring.     | -            | -               | -             | -          |
| <b>PC3.</b> Utilize pivot tables and "What-If" analysis for insights.                        | -            | -               | -             | -          |
| <i>Database Management</i>                                                                   | -            | -               | -             | -          |
| <b>PC4.</b> Understand database concepts and implement SQL queries for data retrieval        | -            | -               | -             | -          |
| <b>PC5.</b> Ensure data integrity using keys and constraints.                                | -            | -               | -             | -          |
| <b>PC6.</b> Perform data migration and aggregation operations.                               | -            | -               | -             | -          |
| <i>Statistical Analysis and Data Interpretation</i>                                          | -            | -               | -             | -          |
| <b>PC7.</b> Implement statistical techniques, including probability and regression analysis. | -            | -               | -             | -          |
| <b>PC8.</b> Conduct descriptive analytics for trend identification.                          | -            | -               | -             | -          |
| <b>PC9.</b> Utilize statistical models for decision-making.                                  | -            | -               | -             | -          |
| <i>Open-Source Data Management</i>                                                           | -            | -               | -             | -          |
| <b>PC10.</b> Work with Linux-based systems for file management and command-line operations.  | -            | -               | -             | -          |
| <b>PC11.</b> Utilize scripting for data handling and automation.                             | -            | -               | -             | -          |
| <i>Data Visualization and Reporting</i>                                                      | -            | -               | -             | -          |
| <b>PC12.</b> Use Tableau to create interactive dashboards and reports.                       | -            | -               | -             | -          |

### Qualification Pack

| Assessment Criteria for Outcomes                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> Connect and analyze data from multiple sources.                     | -            | -               | -             | -          |
| <b>PC14.</b> Generate graphical representations to enhance data storytelling.    | -            | -               | -             | -          |
| <i>Implementation of Real-World Use Cases</i>                                    | -            | -               | -             | -          |
| <b>PC15.</b> Apply acquired skills to industry-relevant datasets.                | -            | -               | -             | -          |
| <b>PC16.</b> Develop reports with actionable insights.                           | -            | -               | -             | -          |
| <b>PC17.</b> Present findings using visual and statistical data interpretations. | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                 | -            | -               | -             | <b>30</b>  |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                   |
|----------------------------|---------------------------------------------------|
| <b>NOS Code</b>            | NIE/ITS/N1116                                     |
| <b>NOS Name</b>            | Implementation of Data Analysis Assistant Project |
| <b>Sector</b>              | Information Technology Sector                     |
| <b>Sub-Sector</b>          |                                                   |
| <b>Occupation</b>          | DATA ANALYTICS                                    |
| <b>NSQF Level</b>          | 3                                                 |
| <b>Credits</b>             | 1                                                 |
| <b>Version</b>             | 1.0                                               |
| <b>Last Reviewed Date</b>  | 29/03/2023                                        |
| <b>Next Review Date</b>    | 29/03/2026                                        |
| <b>NSQC Clearance Date</b> | 29/03/2023                                        |

## Qualification Pack

### NIE/ITS/N1105: Visualize data graphically

#### Description

Introduction to Tableau involves understanding its data visualization capabilities and functionalities. Connecting Tableau to various data sources such as Excel, CSV, and databases allows for seamless data integration. Analyzing data using Tableau's robust features, including formatting options, sorting capabilities, and diverse charting tools, enables users to gain valuable insights and present data in a visually appealing and comprehensive manner.

#### Scope

The scope covers the following :

- The scope encompasses a comprehensive understanding of Tableau, focusing on its data visualization capabilities and the process of connecting to diverse data sources such as Excel, CSV, and databases. It includes extensive training in data analysis through Tableau's advanced features, encompassing formatting techniques, sorting functionalities, and various charting tools, enabling users to effectively transform complex datasets into insightful visual representations for informed decision-making and data presentation

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Familiarization with Tableau's interface and its capabilities for data visualization and analysis. Learning the process of linking Tableau to Excel and CSV files for seamless data integration and analysis
- PC2.** Understanding how to establish connections between Tableau and various databases to access and analyze data directly from these sources, Exploring Tableau's data analysis tools and functionalities for dissecting and understanding complex datasets
- PC3.** Mastering formatting techniques within Tableau to enhance the visual appeal of data visualizations and dashboards. Understanding the sorting capabilities in Tableau to arrange data in a meaningful and logical order for effective analysis. Utilizing Tableau's diverse charting tools to create various types of visual representations, including bar charts, line graphs, and pie charts, for comprehensive data presentation.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Introduction to Tableau and Connecting to Excel, CSV Text Files
- KU2.** Connecting to Databases and its analysis
- KU3.** Formatting, Sorting and Charts

#### Generic Skills (GS)

User/individual on the job needs to know how to:



## Qualification Pack

### **GS1.** • Organizational Skills:

- The effective management of the tasks outlined requires a set of crucial organizational skills, including proficient time management for efficient utilization of time during the learning process and the establishment of connections to various data sources. Moreover, meticulous task prioritization is essential for understanding the importance of each step, from data connection to analysis and visualization, while maintaining a systematic approach to data organization within the Tableau platform. Additionally, attention to detail plays a key role in ensuring the accurate representation of data through appropriate formatting, sorting, and chart selection, fostering a coherent and insightful data visualization process

### **GS2.** • Professional Skills:

- Proficiency in the tasks outlined demands a combination of professional skills, including a comprehensive understanding of data visualization principles and the ability to effectively communicate complex data insights. Proficient technical expertise in Tableau and data connection management is essential for streamlined data integration and analysis. Critical thinking skills are crucial for in-depth data analysis and the selection of appropriate data visualization tools, while attention to detail ensures accurate and insightful data presentation. Effective problem-solving abilities play a vital role in overcoming any technical challenges, fostering a seamless and efficient data analysis and visualization process.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 32           | 14              | -             | -          |
| <b>PC1.</b> Familiarization with Tableau's interface and its capabilities for data visualization and analysis. Learning the process of linking Tableau to Excel and CSV files for seamless data integration and analysis                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <b>PC2.</b> Understanding how to establish connections between Tableau and various databases to access and analyze data directly from these sources, Exploring Tableau's data analysis tools and functionalities for dissecting and understanding complex datasets                                                                                                                                                                               | -            | -               | -             | -          |
| <b>PC3.</b> Mastering formatting techniques within Tableau to enhance the visual appeal of data visualizations and dashboards. Understanding the sorting capabilities in Tableau to arrange data in a meaningful and logical order for effective analysis. Utilizing Tableau's diverse charting tools to create various types of visual representations, including bar charts, line graphs, and pie charts, for comprehensive data presentation. | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>32</b>    | <b>14</b>       | <b>-</b>      | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                               |
|----------------------------|-------------------------------|
| <b>NOS Code</b>            | NIE/ITS/N1105                 |
| <b>NOS Name</b>            | Visualize data graphically    |
| <b>Sector</b>              | Information Technology Sector |
| <b>Sub-Sector</b>          | Data Science & Engineering    |
| <b>Occupation</b>          | Data Analytics                |
| <b>NSQF Level</b>          | 4                             |
| <b>Credits</b>             | 2                             |
| <b>Version</b>             | 1.0                           |
| <b>Last Reviewed Date</b>  | 29/03/2023                    |
| <b>Next Review Date</b>    | 29/03/2026                    |
| <b>NSQC Clearance Date</b> | 29/03/2023                    |

## Qualification Pack

### NIE/ITS/N1105: Perform basic calculation using spreadsheet-

#### Description

Spreadsheet software facilitates the creation, organization, and storage of multiple worksheets within workbooks, enabling users to manage data efficiently. It also allows for the customization of text formats and alignment, supports basic functions and formulas for data manipulation, enables sorting for data organization, and facilitates the creation of graphs for visual data representation and analysis.

#### Scope

The scope covers the following :

- Spreadsheet features, Creating and saving worksheet and workbook layouts, learning basic functions and formulas, sorting and Graphs

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.**
- Creating and saving worksheet and workbook:
  - Understanding of file management and organization in a spreadsheet environment.
  - Proficiency in creating, saving, and managing multiple worksheets within a workbook.
- PC2.**
- Knowledge of formatting tools to enhance data presentation and readability.
  - Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.
- PC3.**
- Proficiency in using basic mathematical and logical functions for data manipulation.
  - Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet.
  - Understanding the importance of data organization and its impact on data analysis.
  - Proficiency in sorting data based on specific criteria for effective data management and interpretation.
- PC4.**
- Ability to interpret and present data visually through graphs and charts.
  - Understanding the use of graphical representations for effective communication and data analysis.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the importance of data organization and its impact on data analysis. Proficiency in sorting data based on specific criteria for effective data management and interpretation.
- KU2.** Proficiency in using basic mathematical and logical functions for data manipulation. Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet.
- KU3.** Knowledge of formatting tools to enhance data presentation and readability. Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.

## Qualification Pack

- KU4.** Understanding of file management and organization in a spreadsheet environment.  
Proficiency in creating, saving, and managing multiple worksheets within a workbook.

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- Organizational Skills:
    - Effective management of data, files, and workbooks.
  - Analytical Skills: Ability to analyze and interpret data using functions and formulas.
  - Attention to Detail: Ensuring accurate data entry, formatting, and alignment.
  - Problem-Solving: Identifying and resolving issues related to data organization and presentation.
  - Communication Skills: Effectively conveying insights through well-designed graphs and charts.
- GS2.**
- Professional Skills:
    - Data Management: Efficiently handling large datasets and maintaining data integrity.
  - Technical Proficiency: Mastery of spreadsheet software tools and functions for optimized usage.
  - Presentation Skills: Creating visually appealing and informative data presentations for professional settings.
  - Critical Thinking: Evaluating data trends and patterns to make informed business decisions.
  - Collaboration: Working with teams to present and analyze data effectively for reports.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>25</b>    | <b>11</b>       | -             | -          |
| <b>PC1.</b> <ul style="list-style-type: none"> <li>• Creating and saving worksheet and workbook:</li> <li>• Understanding of file management and organization in a spreadsheet environment.</li> <li>• Proficiency in creating, saving, and managing multiple worksheets within a workbook.</li> </ul>                                                                                                                                                                              | -            | -               | -             | -          |
| <b>PC2.</b> <ul style="list-style-type: none"> <li>• Knowledge of formatting tools to enhance data presentation and readability.</li> <li>• Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.</li> </ul>                                                                                                                                                                                                       | -            | -               | -             | -          |
| <b>PC3.</b> <ul style="list-style-type: none"> <li>• Proficiency in using basic mathematical and logical functions for data manipulation.</li> <li>• Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet.</li> <li>• Understanding the importance of data organization and its impact on data analysis.</li> <li>• Proficiency in sorting data based on specific criteria for effective data management and interpretation.</li> </ul> | -            | -               | -             | -          |
| <b>PC4.</b> <ul style="list-style-type: none"> <li>• Ability to interpret and present data visually through graphs and charts.</li> <li>• Understanding the use of graphical representations for effective communication and data analysis.</li> </ul>                                                                                                                                                                                                                              | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>25</b>    | <b>11</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                         |                                              |
|-------------------------|----------------------------------------------|
| <b>NOS Code</b>         | NIE/ITS/N1105                                |
| <b>NOS Name</b>         | Perform basic calculation using spreadsheet- |
| <b>Sector</b>           | Information Technology Sector                |
| <b>Sub-Sector</b>       | Data Science & Engineering                   |
| <b>Occupation</b>       | Data Analytics                               |
| <b>NSQF Level</b>       | 3                                            |
| <b>Credits</b>          | 1                                            |
| <b>Version</b>          | 1.0                                          |
| <b>Next Review Date</b> | NA                                           |

## Qualification Pack

### NIE/ITS/N1109: Implementation of use cases of Data Analytics

#### Description

The implementation of use cases in data analytics involves a systematic approach to extracting insights and value from large datasets, typically to address specific business challenges or opportunities. It starts with defining the business objectives and identifying the relevant data sources, followed by data collection, processing, and preparation for analysis. Various data analytics techniques, such as descriptive, diagnostic, predictive, and prescriptive analytics, are employed to uncover patterns, trends, and correlations within the data. Advanced tools and technologies like data mining, machine learning, and artificial intelligence are often utilized to perform complex analyses and generate actionable insights.

#### Scope

The scope covers the following :

- Implementation of use cases in data analytics encompasses a comprehensive understanding of the entire data analytics process, including data collection, processing, analysis, and interpretation. It also includes proficiency in a variety of data analytics techniques, tools, and technologies, such as descriptive, diagnostic, predictive, and prescriptive analytics, as well as data mining, machine learning, and artificial intelligence.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

**PC1.** The implementation of use cases in data analytics involve several key aspects. Firstly, ensuring alignment between the data analytics process and the defined business objectives is crucial, as it facilitates the extraction of meaningful insights that can drive tangible business outcomes. Efficient data collection, processing, and preparation, with an emphasis on data quality and integrity, are essential for accurate and reliable analysis. Implementing a range of data analytics techniques and lev

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** Use cases in data analytics require a comprehensive grasp of various data analytics methodologies, including descriptive, diagnostic, predictive, and prescriptive analytics, along with proficiency in data collection, processing, and analysis techniques. It also involves a deep understanding of data mining, machine learning, and artificial intelligence algorithms, and their practical application in deriving actionable insights from complex datasets. Additionally, a strong command of statistical a

#### Generic Skills (GS)

User/individual on the job needs to know how to:



## Qualification Pack

**GS1.** Candidate will encompass expertise in utilizing various data analytics tools and technologies, such as statistical software (e.g., R, Python), database management systems, and business intelligence platforms. Additionally, skills in data mining, machine learning, and predictive modeling are crucial. A strong understanding of industry-specific knowledge and business acumen, along with the ability to translate analytical insights into actionable business strategies, is also important

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 16           | 7               | -             | -          |
| <b>PC1.</b> The implementation of use cases in data analytics involve several key aspects. Firstly, ensuring alignment between the data analytics process and the defined business objectives is crucial, as it facilitates the extraction of meaningful insights that can drive tangible business outcomes. Efficient data collection, processing, and preparation, with an emphasis on data quality and integrity, are essential for accurate and reliable analysis. Implementing a range of data analytics techniques and lev | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>16</b>    | <b>7</b>        | <b>-</b>      | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                               |
|----------------------------|-----------------------------------------------|
| <b>NOS Code</b>            | NIE/ITS/N1109                                 |
| <b>NOS Name</b>            | Implementation of use cases of Data Analytics |
| <b>Sector</b>              | Information Technology Sector                 |
| <b>Sub-Sector</b>          | Data Science & Engineering                    |
| <b>Occupation</b>          | Data Analytics                                |
| <b>NSQF Level</b>          | 4                                             |
| <b>Credits</b>             | 1                                             |
| <b>Version</b>             | 1.0                                           |
| <b>Last Reviewed Date</b>  | 29/03/2023                                    |
| <b>Next Review Date</b>    | 29/03/2026                                    |
| <b>NSQC Clearance Date</b> | 29/03/2023                                    |

## Qualification Pack

### NIE/ITS/N1109: Manage structured data-

#### Description

Introduction to database , advantages of DBMS, Concept of keys: candidate key, primary key, alternate key, Foreign key, Fundamental integrity rules: entity integrity, referential integrity. Entity-relationship model .ER-diagram, SQL, Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations. Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations.

#### Scope

The scope covers the following :

- It encompasses the foundational understanding of databases, the benefits of Database Management Systems (DBMS), and an in-depth exploration of key concepts including candidate keys, primary keys, alternate keys, foreign keys, entity integrity, and referential integrity. Additionally, it involves the application of the Entity-Relationship model for database design, the mastery of SQL for efficient data management, and a holistic view of Knowledge Discovery in Databases (KDD), data mining techniques, and the architecture of data warehouses. The scope further extends to encompass the methodologies of migrating data to data warehouses, the pivotal process of data cleaning, and the application of aggregation operations to ensure data accuracy and integrity for effective decision-making and analysis.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding the fundamentals of databases, their types, and their importance in information management systems.
- PC2.** Exploring the benefits of Database Management Systems (DBMS) in terms of data organization, retrieval, security, and integrity maintenance. Comprehensive exploration of candidate keys, primary keys, alternate keys, and foreign keys, along with their roles in data integrity and relationship management.
- PC3.** Delving into entity integrity and referential integrity, elucidating their significance in maintaining data consistency and accuracy within the database. Learning the conceptual and practical aspects of the ER model for designing and representing database structures and relationships between entities. Mastering the Structured Query Language (SQL) for efficient data management, retrieval, and manipulation within the database system.
- PC4.**
  - Understanding the processes involved in discovering valuable knowledge from large datasets, emphasizing data analysis, and extraction of useful patterns and insights. Exploring various data mining techniques, algorithms, and tools for uncovering patterns, correlations, and anomalies in large datasets. Comprehending the architecture and design of data warehouses, and their significance in providing a consolidated and centralized repository for decision-making and analysis.
  - Migrating Data to Data

#### Knowledge and Understanding (KU)

## Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Individual learn about Concepts of Database and Its Advantages
- KU2.** Concept of Keys, Fundamental Integrity Rules, Entity-Relationship Model (ER Diagram):
- KU3.** Mastering the Structured Query Language (SQL), Knowledge Discovery in Databases (KDD),
- KU4.** Learning concepts of Data Mining and Data Warehouse

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
  - Time Management: Efficiently managing tasks and adhering to project timelines, especially during data migration and cleaning operations. Resource Management: Effectively utilizing available tools and resources for database management and data mining processes. Data Organization: Structuring data in a systematic manner to ensure efficient access and retrieval during analysis and reporting.
  - Project Planning: Strategically planning and implementing data warehouse initiatives, taking into account data migration, cleaning, and aggregation operations. Detail Orientation: Paying close attention to data details to maintain accuracy and integrity throughout the data management and analysis processes.
- GS2.** • Professional Skills:
  - Technical Proficiency: Demonstrating an advanced understanding of database management systems, SQL querying, and data mining techniques for proficient handling and analysis of large datasets.
  - Problem-Solving: Resolving complex database-related issues and challenges during data migration, cleaning, and aggregation, ensuring optimal data quality and integrity.
  - Communication Skills: Articulating complex technical concepts and insights to both technical and non-technical stakeholders in a clear and concise manner. Collaboration: Working effectively with multidisciplinary teams, including data analysts, IT professionals, and stakeholders, to ensure seamless data management and analysis processes. Critical Thinking: Evaluating data trends and patterns critically to make informed decisions during data mining and knowledge discovery processes.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 25           | 11              | -             | -          |
| <b>PC1.</b> Understanding the fundamentals of databases, their types, and their importance in information management systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -            | -               | -             | -          |
| <b>PC2.</b> Exploring the benefits of Database Management Systems (DBMS) in terms of data organization, retrieval, security, and integrity maintenance. Comprehensive exploration of candidate keys, primary keys, alternate keys, and foreign keys, along with their roles in data integrity and relationship management.                                                                                                                                                                                                                                                                          | -            | -               | -             | -          |
| <b>PC3.</b> Delving into entity integrity and referential integrity, elucidating their significance in maintaining data consistency and accuracy within the database. Learning the conceptual and practical aspects of the ER model for designing and representing database structures and relationships between entities. Mastering the Structured Query Language (SQL) for efficient data management, retrieval, and manipulation within the database system.                                                                                                                                     | -            | -               | -             | -          |
| <b>PC4.</b> <ul style="list-style-type: none"> <li>Understanding the processes involved in discovering valuable knowledge from large datasets, emphasizing data analysis, and extraction of useful patterns and insights. Exploring various data mining techniques, algorithms, and tools for uncovering patterns, correlations, and anomalies in large datasets.</li> <li>Comprehending the architecture and design of data warehouses, and their significance in providing a consolidated and centralized repository for decision-making and analysis.</li> <li>Migrating Data to Data</li> </ul> | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>25</b>    | <b>11</b>       | <b>-</b>      | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                         |                               |
|-------------------------|-------------------------------|
| <b>NOS Code</b>         | NIE/ITS/N1109                 |
| <b>NOS Name</b>         | Manage structured data-       |
| <b>Sector</b>           | Information Technology Sector |
| <b>Sub-Sector</b>       | Data Science & Engineering    |
| <b>Occupation</b>       | Data Analytics                |
| <b>NSQF Level</b>       | 3                             |
| <b>Credits</b>          | 1                             |
| <b>Version</b>          | 1.0                           |
| <b>Next Review Date</b> | NA                            |

## Qualification Pack

### NIE/ITS/N1112: Analyze data using spreadsheet tool-

#### Description

Student will learn the key elements of data manipulation and analysis within spreadsheet software. Data entails the information input, while organizing it into tables and formatting involves structuring and visually enhancing the presentation. Conditional formatting enables users to dynamically format cells based on specific conditions, advanced charting aids in creating comprehensive visual representations, and pivot tables provide an effective means of data summarization. Data validation ensures the accuracy and reliability of data input, while filtering allows for focused data analysis by selectively displaying subsets of information that meet specified criteria.

#### Scope

The scope covers the following :

- The scope involves comprehensive training in data handling within spreadsheets, focusing on data input, organization, and presentation through tables and formatting. It also includes an in-depth understanding of conditional formatting techniques, advanced charting methodologies for sophisticated data visualization, and the proficient use of pivot tables for efficient data summarization and analysis. Additionally, the scope entails expertise in implementing data validation measures to ensure data accuracy, as well as utilizing filtering techniques to extract specific data subsets, thereby facilitating effective data analysis and decision-making.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding data entry and storage within spreadsheets. Learning to create and manage tables for organized data representation. Mastering formatting tools to enhance the visual presentation of data.
- PC2.** Exploring the application of formatting rules based on specific conditions. Understanding how to dynamically highlight and format data for easy analysis. Learning to use color scales, data bars, and icon sets to represent data variations effectively.
- PC3.** Gaining proficiency in creating a variety of complex charts to visualize data trends and patterns. Understanding the use of trendlines, dual axes, and custom data labels for comprehensive data representation. Learning to create combination charts to display multiple data sets in a single graph. Learning how to summarize and analyze large datasets efficiently. Understanding the process of data aggregation and dynamic restructuring for insightful analysis.
- PC4.** Implementing data validation rules to ensure accurate and consistent data entry. Understanding the use of input messages and error alerts for guiding data input. Learning to restrict data input based on predefined criteria for data integrity. Understanding how to filter data based on specific criteria to focus on relevant information. Exploring the use of filter views to create customized views of data subsets. Learning to apply multiple filters for complex data analysis and visualization.

#### Knowledge and Understanding (KU)



## Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Candidate will gain knowledge of Data manipulation inside ant Table, Formatting and conditional formatting
- KU2.** Exploring the application of formatting rules based on specific conditions. Understanding how to dynamically highlight and format data for easy analysis.
- KU3.** Candidate will gain knowledge of Advance Charting and use of Pivot Table
- KU4.** Implementing data validation rules to ensure accurate and consistent data entry. Understanding the use of input messages and error alerts for guiding data input. Exploring the use of filter views to create customized views of data subsets.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
  - he proficient application of the aforementioned spreadsheet functionalities necessitates strong organizational skills, including adept time management for efficient handling of data tasks, meticulous attention to detail to ensure accurate data organization and analysis, and effective task prioritization to manage various aspects of data manipulation. Additionally, the ability to strategically plan and execute data management projects, coupled with proficient problem-solving capabilities and clear communication skills for conveying data insights, are crucial for successful implementation and interpretation of spreadsheet data
- GS2.** • Professional Skills:
  - Proficient application of these features in a professional context necessitates a strong data analysis proficiency, along with technical expertise in spreadsheet software and related tools for efficient data manipulation. Critical thinking skills are essential for insightful data interpretation, while maintaining data integrity through meticulous validation and filtering procedures is crucial. Effective problem-solving abilities enable the resolution of complex data-related challenges, and adaptability to new techniques ensures the incorporation of the latest advancements in spreadsheet technology. Additionally, collaborative aptitude facilitates effective teamwork in analyzing and interpreting data for informed decision-making and problem-solving in a professional environment.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 50           | 22              | -             | -          |
| <b>PC1.</b> Understanding data entry and storage within spreadsheets. Learning to create and manage tables for organized data representation. Mastering formatting tools to enhance the visual presentation of data.                                                                                                                                                                                                                                                                                                       | -            | -               | -             | -          |
| <b>PC2.</b> Exploring the application of formatting rules based on specific conditions. Understanding how to dynamically highlight and format data for easy analysis. Learning to use color scales, data bars, and icon sets to represent data variations effectively.                                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC3.</b> Gaining proficiency in creating a variety of complex charts to visualize data trends and patterns. Understanding the use of trendlines, dual axes, and custom data labels for comprehensive data representation. Learning to create combination charts to display multiple data sets in a single graph. Learning how to summarize and analyze large datasets efficiently. Understanding the process of data aggregation and dynamic restructuring for insightful analysis.                                     | -            | -               | -             | -          |
| <b>PC4.</b> Implementing data validation rules to ensure accurate and consistent data entry. Understanding the use of input messages and error alerts for guiding data input. Learning to restrict data input based on predefined criteria for data integrity. Understanding how to filter data based on specific criteria to focus on relevant information. Exploring the use of filter views to create customized views of data subsets. Learning to apply multiple filters for complex data analysis and visualization. | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>50</b>    | <b>22</b>       | <b>-</b>      | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                      |
|----------------------------|--------------------------------------|
| <b>NOS Code</b>            | NIE/ITS/N1112                        |
| <b>NOS Name</b>            | Analyze data using spreadsheet tool- |
| <b>Sector</b>              | Information Technology Sector        |
| <b>Sub-Sector</b>          | Data Science & Engineering           |
| <b>Occupation</b>          | Data Analytics                       |
| <b>NSQF Level</b>          | 3                                    |
| <b>Credits</b>             | 2                                    |
| <b>Version</b>             | 1.0                                  |
| <b>Last Reviewed Date</b>  | 29/03/2023                           |
| <b>Next Review Date</b>    | 29/03/2026                           |
| <b>NSQC Clearance Date</b> | 29/03/2023                           |

## Qualification Pack

### NIE/ITS/N1113: Manage data in Open source tool-

#### Description

It is an introduction to Virtual Machines, covering the creation and configuration processes. It further delves into the history and benefits of Linux, exploring the various Linux distributions and focusing on Ubuntu. The scope includes a step-by-step guide to installing Ubuntu, familiarizing users with startup procedures, logging in, and navigating the desktop environment. Additionally, it involves an in-depth understanding of the Linux file system, covering concepts such as file management, real and virtual files, mounting, file searches, file size, and viewing text files. Moreover, the scope encompasses practical guidance on using command-line text editors, creating, searching, and comparing text files, as well as managing files through copying, moving, and other relevant operations within the Linux environment.

#### Scope

The scope covers the following :

- Comprehensive understanding of virtualization concepts with a focus on Virtual Machines, covering their introduction, creation, and configuration. Additionally, it entails an exploration of the history of Linux, its benefits, and the various Linux distributions, with a specific emphasis on Ubuntu. The scope involves a detailed walkthrough of the installation process for Ubuntu, guiding users through startup procedures, logging in, and desktop navigation. It also encompasses an in-depth exploration of the Linux file system, covering fundamental concepts such as file management, file system structure, and various file operations. Furthermore, the scope includes practical training in using command-line text editors, file searching, comparison, and manipulation, enabling users to effectively manage files and directories within the Linux environment.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

**PC1.** Introduction to Virtual Machine, Creating and Configuring Virtual Machine, Linux History

**PC2.** Benefits of Linux, Different Flavors of Linux, Introducing Ubuntu

**PC3.** Installing Ubuntu: Starting Up, Logging in, Exploring the Desktop, Ubuntu Basics

**PC4.** Browsing the File System, Understanding File System Concept, Managing Files: Real and Virtual Files, Mounting, File Searches, File Size, File Space Viewing, Text Files Viewing, Using a Command, Line Text Editor, Creating Files, Searching through Files, Comparing Text Files, Copying, Moving, Managing Files

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Virtual Machines: Knowledge of virtualization technology and software tools for creating and configuring virtual machines. Linux History and Benefits: Understanding the origins, development, and advantages of the Linux operating system over other operating systems.
- KU2.** Linux Distributions: Awareness of various Linux distributions, their features, and use cases, with a specific focus on Ubuntu.

## Qualification Pack

- KU3.** Ubuntu Installation and Basics: Familiarity with the installation process and basic operations within the Ubuntu Linux environment, including desktop navigation and file system management.
- KU4.** File System Operations: Understanding the structure and management of file systems in Linux, including file browsing, searching, size estimation, and text file manipulation using command-line text editors. File Management: Knowledge of file management operations such as copying, moving, and comparing files, as well as handling real and virtual files and mounting external file systems.

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
- The proficient execution of the outlined tasks necessitates adept organizational skills, including effective time management for efficient handling of tasks such as virtual machine configuration, Linux installation, and file management within the Ubuntu environment. Furthermore, meticulous attention to detail is crucial during file system navigation, file searches, and text file manipulation, ensuring accurate and precise data handling. Additionally, proficient task prioritization and problem-solving abilities are essential for addressing any challenges that may arise during the installation process and for maintaining an organized approach to data and file management operations.
- GS2.** • Professional Skills:
- Proficiency in the aforementioned tasks requires a combination of professional skills, including a strong grasp of virtualization technologies and Linux operating systems, effective problem-solving abilities to address technical challenges during virtual machine configuration and Linux installation, and meticulous attention to detail for accurate file system management and file manipulation. Additionally, effective communication skills are essential for documenting procedures and troubleshooting steps, while adaptability and continuous learning ensure the ability to stay updated with the latest developments in virtualization and Linux environments.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                                                                                                                                 | <b>25</b>    | <b>11</b>       | -             | -          |
| <b>PC1.</b> Introduction to Virtual Machine, Creating and Configuring Virtual Machine, Linux History                                                                                                                                                                                                                            | -            | -               | -             | -          |
| <b>PC2.</b> Benefits of Linux, Different Flavors of Linux, Introducing Ubuntu                                                                                                                                                                                                                                                   | -            | -               | -             | -          |
| <b>PC3.</b> Installing Ubuntu: Starting Up, Logging in, Exploring the Desktop, Ubuntu Basics                                                                                                                                                                                                                                    | -            | -               | -             | -          |
| <b>PC4.</b> Browsing the File System, Understanding File System Concept, Managing Files: Real and Virtual Files, Mounting, File Searches, File Size, File Space Viewing, Text Files Viewing, Using a Command , Line Text Editor, Creating Files, Searching through Files, Comparing Text Files, Copying, Moving, Managing Files | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                                                                                | <b>25</b>    | <b>11</b>       | -             | -          |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                  |
|----------------------------|----------------------------------|
| <b>NOS Code</b>            | NIE/ITS/N1113                    |
| <b>NOS Name</b>            | Manage data in Open source tool- |
| <b>Sector</b>              | Information Technology Sector    |
| <b>Sub-Sector</b>          | Data Science & Engineering       |
| <b>Occupation</b>          | Data Analytics                   |
| <b>NSQF Level</b>          | 3                                |
| <b>Credits</b>             | 1                                |
| <b>Version</b>             | 1.0                              |
| <b>Last Reviewed Date</b>  | 29/03/2023                       |
| <b>Next Review Date</b>    | 29/03/2026                       |
| <b>NSQC Clearance Date</b> | 29/03/2023                       |

## Qualification Pack

### NIE/ITS/N1114: Visualize Data Graphically

#### Description

Spreadsheet software facilitates the creation, organization, and storage of multiple worksheets within workbooks, enabling users to manage data efficiently. It also allows for the customization of text formats and alignment, supports basic functions and formulas for data manipulation, enables sorting for data organization, and facilitates the creation of graphs for visual data representation and analysis.

#### Scope

The scope covers the following :

- Spreadsheet features, Creating and saving worksheet and workbook layouts, learning basic functions and formulas, sorting and Graphs

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.**
- Creating and saving worksheet and workbook:
  - Understanding of file management and organization in a spreadsheet environment.
  - Proficiency in creating, saving, and managing multiple worksheets within a workbook.
- PC2.**
- Knowledge of formatting tools to enhance data presentation and readability.
  - Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.
- PC3.**
- Proficiency in using basic mathematical and logical functions for data manipulation.
  - Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet.
  - Understanding the importance of data organization and its impact on data analysis.
  - Proficiency in sorting data based on specific criteria for effective data management and interpretation.
- PC4.**
- Ability to interpret and present data visually through graphs and charts.
  - Understanding the use of graphical representations for effective communication and data analysis.

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the importance of data organization and its impact on data analysis. Proficiency in sorting data based on specific criteria for effective data management and interpretation.
- KU2.** Proficiency in using basic mathematical and logical functions for data manipulation. Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet.
- KU3.** Knowledge of formatting tools to enhance data presentation and readability. Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.



## Qualification Pack

- KU4.** Understanding of file management and organization in a spreadsheet environment.  
Proficiency in creating, saving, and managing multiple worksheets within a workbook.

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:  
• Effective management of data, files, and workbooks. Analytical Skills: Ability to analyze and interpret data using functions and formulas. Attention to Detail: Ensuring accurate data entry, formatting, and alignment. Problem-Solving: Identifying and resolving issues related to data organization and presentation. Communication Skills: Effectively conveying insights through well-designed graphs and charts.
- GS2.** • Professional Skills:  
• Data Management: Efficiently handling large datasets and maintaining data integrity. Technical Proficiency: Mastery of spreadsheet software tools and functions for optimized usage. Presentation Skills: Creating visually appealing and informative data presentations for professional settings. Critical Thinking: Evaluating data trends and patterns to make informed business decisions. Collaboration: Working with teams to present and analyze data effectively for reports.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                            | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 50           | 22              | -             | -          |
| <b>PC1.</b> <ul style="list-style-type: none"> <li>Creating and saving worksheet and workbook:</li> <li>Understanding of file management and organization in a spreadsheet environment.</li> <li>Proficiency in creating, saving, and managing multiple worksheets within a workbook.</li> </ul>                                                                                                                                                                            | -            | -               | -             | -          |
| <b>PC2.</b> <ul style="list-style-type: none"> <li>Knowledge of formatting tools to enhance data presentation and readability.</li> <li>Ability to utilize text alignment and formatting options to create visually appealing and comprehensible spreadsheets.</li> </ul>                                                                                                                                                                                                   | -            | -               | -             | -          |
| <b>PC3.</b> <ul style="list-style-type: none"> <li>Proficiency in using basic mathematical and logical functions for data manipulation.</li> <li>Understanding of how to apply formulas to perform calculations and analysis within a spreadsheet.</li> <li>Understanding the importance of data organization and its impact on data analysis.</li> <li>Proficiency in sorting data based on specific criteria for effective data management and interpretation.</li> </ul> | -            | -               | -             | -          |
| <b>PC4.</b> <ul style="list-style-type: none"> <li>Ability to interpret and present data visually through graphs and charts.</li> <li>Understanding the use of graphical representations for effective communication and data analysis.</li> </ul>                                                                                                                                                                                                                          | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>50</b>    | <b>22</b>       | <b>-</b>      | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                               |
|----------------------------|-------------------------------|
| <b>NOS Code</b>            | NIE/ITS/N1114                 |
| <b>NOS Name</b>            | Visualize Data Graphically    |
| <b>Sector</b>              | Information Technology Sector |
| <b>Sub-Sector</b>          | Data Science & Engineering    |
| <b>Occupation</b>          | Data Analytics                |
| <b>NSQF Level</b>          | 3                             |
| <b>Credits</b>             | 2                             |
| <b>Version</b>             | 1.0                           |
| <b>Last Reviewed Date</b>  | 29/03/2023                    |
| <b>Next Review Date</b>    | 29/03/2026                    |
| <b>NSQC Clearance Date</b> | 29/03/2023                    |

## Qualification Pack

### NIE/ITS/N1115: Implementation of use cases of Data Analytics-

#### Description

Introduction to database , advantages of DBMS, Concept of keys: candidate key, primary key, alternate key, Foreign key, Fundamental integrity rules: entity integrity, referential integrity. Entity-relationship model .ER-diagram, SQL, Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations. Knowledge Discovery in Databases, Data Mining, Data warehouse. Migrating data from source to data warehouse, cleaning, aggregation operations.

#### Scope

The scope covers the following :

- It encompasses the foundational understanding of databases, the benefits of Database Management Systems (DBMS), and an in-depth exploration of key concepts including candidate keys, primary keys, alternate keys, foreign keys, entity integrity, and referential integrity. Additionally, it involves the application of the Entity-Relationship model for database design, the mastery of SQL for efficient data management, and a holistic view of Knowledge Discovery in Databases (KDD), data mining techniques, and the architecture of data warehouses. The scope further extends to encompass the methodologies of migrating data to data warehouses, the pivotal process of data cleaning, and the application of aggregation operations to ensure data accuracy and integrity for effective decision-making and analysis.

#### Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding the fundamentals of databases, their types, and their importance in information management systems.
- PC2.** Exploring the benefits of Database Management Systems (DBMS) in terms of data organization, retrieval, security, and integrity maintenance. Comprehensive exploration of candidate keys, primary keys, alternate keys, and foreign keys, along with their roles in data integrity and relationship management.
- PC3.** Delving into entity integrity and referential integrity, elucidating their significance in maintaining data consistency and accuracy within the database. Learning the conceptual and practical aspects of the ER model for designing and representing database structures and relationships between entities. Mastering the Structured Query Language (SQL) for efficient data management, retrieval, and manipulation within the database system.
- PC4.**
  - Understanding the processes involved in discovering valuable knowledge from large datasets, emphasizing data analysis, and extraction of useful patterns and insights. Exploring various data mining techniques, algorithms, and tools for uncovering patterns, correlations, and anomalies in large datasets. Comprehending the architecture and design of data warehouses, and their significance in providing a consolidated and centralized repository for decision-making and analysis.
  - Migrating Data to Data

#### Knowledge and Understanding (KU)

## Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Individual learn about Concepts of Database and Its Advantages
- KU2.** Concept of Keys, Fundamental Integrity Rules, Entity-Relationship Model (ER Diagram):
- KU3.** Mastering the Structured Query Language (SQL), Knowledge Discovery in Databases (KDD),
- KU4.** Learning concepts of Data Mining and Data Warehouse

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Organizational Skills:
  - Time Management: Efficiently managing tasks and adhering to project timelines, especially during data migration and cleaning operations. Resource Management: Effectively utilizing available tools and resources for database management and data mining processes. Data Organization: Structuring data in a systematic manner to ensure efficient access and retrieval during analysis and reporting.
  - Project Planning: Strategically planning and implementing data warehouse initiatives, taking into account data migration, cleaning, and aggregation operations. Detail Orientation: Paying close attention to data details to maintain accuracy and integrity throughout the data management and analysis processes.
- GS2.** • Professional Skills:
  - Technical Proficiency: Demonstrating an advanced understanding of database management systems, SQL querying, and data mining techniques for proficient handling and analysis of large datasets.
  - Problem-Solving: Resolving complex database-related issues and challenges during data migration, cleaning, and aggregation, ensuring optimal data quality and integrity.
  - Communication Skills: Articulating complex technical concepts and insights to both technical and non-technical stakeholders in a clear and concise manner. Collaboration: Working effectively with multidisciplinary teams, including data analysts, IT professionals, and stakeholders, to ensure seamless data management and analysis processes. Critical Thinking: Evaluating data trends and patterns critically to make informed decisions during data mining and knowledge discovery processes.

## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 25           | 13              | -             | -          |
| <b>PC1.</b> Understanding the fundamentals of databases, their types, and their importance in information management systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -            | -               | -             | -          |
| <b>PC2.</b> Exploring the benefits of Database Management Systems (DBMS) in terms of data organization, retrieval, security, and integrity maintenance. Comprehensive exploration of candidate keys, primary keys, alternate keys, and foreign keys, along with their roles in data integrity and relationship management.                                                                                                                                                                                                                                                                          | -            | -               | -             | -          |
| <b>PC3.</b> Delving into entity integrity and referential integrity, elucidating their significance in maintaining data consistency and accuracy within the database. Learning the conceptual and practical aspects of the ER model for designing and representing database structures and relationships between entities. Mastering the Structured Query Language (SQL) for efficient data management, retrieval, and manipulation within the database system.                                                                                                                                     | -            | -               | -             | -          |
| <b>PC4.</b> <ul style="list-style-type: none"> <li>Understanding the processes involved in discovering valuable knowledge from large datasets, emphasizing data analysis, and extraction of useful patterns and insights. Exploring various data mining techniques, algorithms, and tools for uncovering patterns, correlations, and anomalies in large datasets.</li> <li>Comprehending the architecture and design of data warehouses, and their significance in providing a consolidated and centralized repository for decision-making and analysis.</li> <li>Migrating Data to Data</li> </ul> | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>25</b>    | <b>13</b>       | <b>-</b>      | <b>-</b>   |

## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                |
|----------------------------|------------------------------------------------|
| <b>NOS Code</b>            | NIE/ITS/N1115                                  |
| <b>NOS Name</b>            | Implementation of use cases of Data Analytics- |
| <b>Sector</b>              | Information Technology Sector                  |
| <b>Sub-Sector</b>          | Data Science & Engineering                     |
| <b>Occupation</b>          | Data Analytics                                 |
| <b>NSQF Level</b>          | 3                                              |
| <b>Credits</b>             | 1                                              |
| <b>Version</b>             | 1.0                                            |
| <b>Last Reviewed Date</b>  | 29/03/2023                                     |
| <b>Next Review Date</b>    | 29/03/2026                                     |
| <b>NSQC Clearance Date</b> | 29/03/2023                                     |

## Assessment Guidelines and Assessment Weightage

### Assessment Guidelines

1. Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria.
2. The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, Internal assessment, Project/Presentation/

Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

## Qualification Pack

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number.
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.
6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.
10. Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

### Minimum Aggregate Passing % at QP Level : 50

**(Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

## Assessment Weightage

Compulsory NOS



### Qualification Pack

| National Occupational Standards                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|-----------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| DGT/VSQ/N0101.Employability Skills (30 Hours)                   | 20           | 30              | -             | -          | 50          | 9          |
| NIE/ITS/N1116.Implementation of Data Analysis Assistant Project | -            | -               | -             | 30         | 30          | 8          |
| NIE/ITS/N1105.Perform basic calculation using spreadsheet-      | 25           | 11              | -             | -          | 36          | 10         |
| NIE/ITS/N1109.Manage structured data-                           | 25           | 11              | -             | -          | 36          | 10         |
| NIE/ITS/N1112.Analyze data using spreadsheet tool-              | 50           | 22              | -             | -          | 72          | 21         |
| NIE/ITS/N1113.Manage data in Open source tool-                  | 25           | 11              | -             | -          | 36          | 10         |
| NIE/ITS/N1114.Visualize Data Graphically                        | 50           | 22              | -             | -          | 72          | 21         |
| NIE/ITS/N1115.Implementation of use cases of Data Analytics-    | 25           | 13              | -             | -          | 38          | 11         |
| <b>Total</b>                                                    | <b>220</b>   | <b>120</b>      | <b>-</b>      | <b>30</b>  | <b>370</b>  | <b>100</b> |

## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |

## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |

## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |